Certainty from Uncertainty: Multi-granularity Labeling Inspired by Quantum Collapse for Learning with Noisy Labels in Fault Diagnosis

Fir Dunkin (邓金), *Graduate Student Member IEEE*, Xinde Li, *Senior Member IEEE*, Zhentong Zhang, Kui Wang, Tianrong Gao, Guoliang Wu, and Zhijun Li, *Fellow, IEEE*

***Abstract*— Deep learning has demonstrated exceptional performance in fault diagnosis tasks that rely on large-scale datasets. However, the high cost of annotating such datasets has led to the emergence of various automatic annotation methods. While these methods reduce labeling costs, they inevitably introduce noisy labels, which pose significant challenges to the generalization and accuracy of deep learning-based diagnostic models. Although Learning with Noisy Labels (LNL) methods mitigate the adverse effects of noisy labels through strategies such as sample separation or label correction, many rely heavily on their own prediction results to guide subsequent training, which often introduces confirmation bias, limiting the effectiveness of the trained models. To address this limitation, this paper draws inspiration from quantum collapse and proposes a novel LNL strategy named Multi-granularity Labeling (*MgL*). By integrating observed labels, pseudo-labels, and collapsed labels, MgL constructs the multi-granularity labels, designed to suppress confirmation bias and improve the model's tolerance to noisy labels. Extensive experiments validate the effectiveness and superiority of MgL, particularly on training datasets with high noise intensity, such as those with 90% symmetric noise. This advancement offers promising opportunities for applying datasets with lower annotation costs in real-world scenarios, ultimately contributing to intelligent diagnostic systems.**

***Index Terms*— Learning with noisy labels, Information fusion, Confirmation bias, Granular computing, Time series classification.**

This work was supported in part by the National Natural Science Foundation of China under Grant 62233003 and 62073072, the Key Projects of Key R&D Program of Jiangsu Province under Grant BE2020006 and BE2020006-1, and the Shenzhen Science and Technology Program under Grant JCYJ20210324132202005 and JCYJ20220818101206014 (*Corresponding author: *Xinde Li*).

Fir Dunkin, Zhentong Zhang, and Guoliang Wu, are with Key Laboratory of Measurement and Control of CSE, School of Automation, Southeast University, Nanjing 210096, China (e-mails: dunkin2fir@ieee.org; zhentong_zhang@seu.edu.cn; 230208677@seu.edu.cn).

Xinde Li is with Key Laboratory of Measurement and Control of CSE, School of Automation, Southeast University, Nanjing 210096, China, also is with Faculty of Robot Science and Engineering, Northeastern University, Shenyang 110167, China, and also is with Southeast University Shenzhen Research Institute, Shenzhen 518063, China. (e-mail: xindeli@seu.edu.cn).

Kui Wang and Tianrong Gao are with North Information Control Group Co., Ltd., Nanjing 211153, China (e-mails: wangkui_sim528@nicgcl.com.cn; gaotianrong_528@nicgcl.com.cn).

Zhijun Li is with the School of Mechanical Engineering, Tongji University, Shanghai 200092, China, and also with Shanghai Key Laboratory of Wearable Robotics and Human-Machine Interaction, Shanghai 201804, China. (e-mail: zjli@ieee.org).

I. INTRODUCTION

IN the era of big data, deep learning-based intelligent fault diagnosis has achieved remarkable advancements in both theoretical research and practical applications [1]. In particular, time-series modeling has emerged as a central yet challenging issue in this field due to the complexity of temporal dependencies and distribution shifts across domains. Recent efforts have explored various strategies to tackle these challenges, such as adversarial Transformer models for robust multivariate anomaly detection [2], support-sample-assisted generalization for domain-agnostic diagnosis [3], and gradient-aligned transfer mechanisms for unsupervised cross-domain adaptation [4].

Despite these advances, most existing approaches heavily rely on large-scale datasets with precise annotations [5]. Acquiring such datasets is both costly and time-consuming, especially for tasks requiring expert knowledge [6], thereby limiting their application in many industrial scenarios.

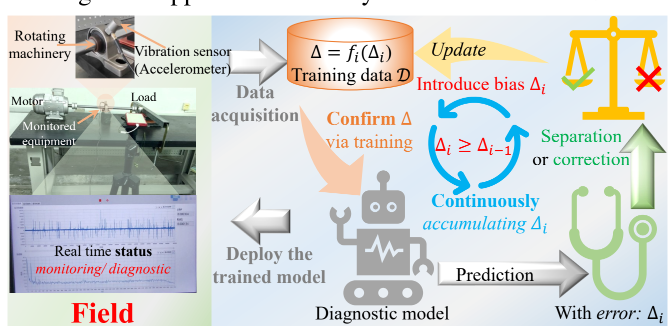


Fig. 1. Confirmation bias in LNL: Most existing methods rely on model predictions during training to refine data, e.g. separating clean samples or correcting noisy labels. However, these predictions inherently carry errors $\boldsymbol{\Delta}_i$ due to the model's limitations. When these error-laden predictions are used to guide subsequent training iterations, the model reinforces these inaccuracies, mistaking them for valid information, and thus undermining its generalization capability.

To address this challenge, alternative labeling strategies such as crowdsourcing [7] and data engines [8] have garnered significant attention due to their cost-effectiveness. Unfortunately, these methods often introduce noisy labels [9], where the observed label y deviates from the true class $\hat{y}$. These discrepancies can be abstracted through an unknown transition matrix $T(\mathcal{Y} \mid \hat{\mathcal{Y}}) = \mathrm{P}\big(\mathcal{Y} = k \mid \hat{\mathcal{Y}} \neq k\big)$. When the noise level becomes substantial, the training process is driven away from the underlying data distribution, leading to misplaced classification boundaries and degraded generalization [10]. As a result, effectively mitigating the detrimental effects of

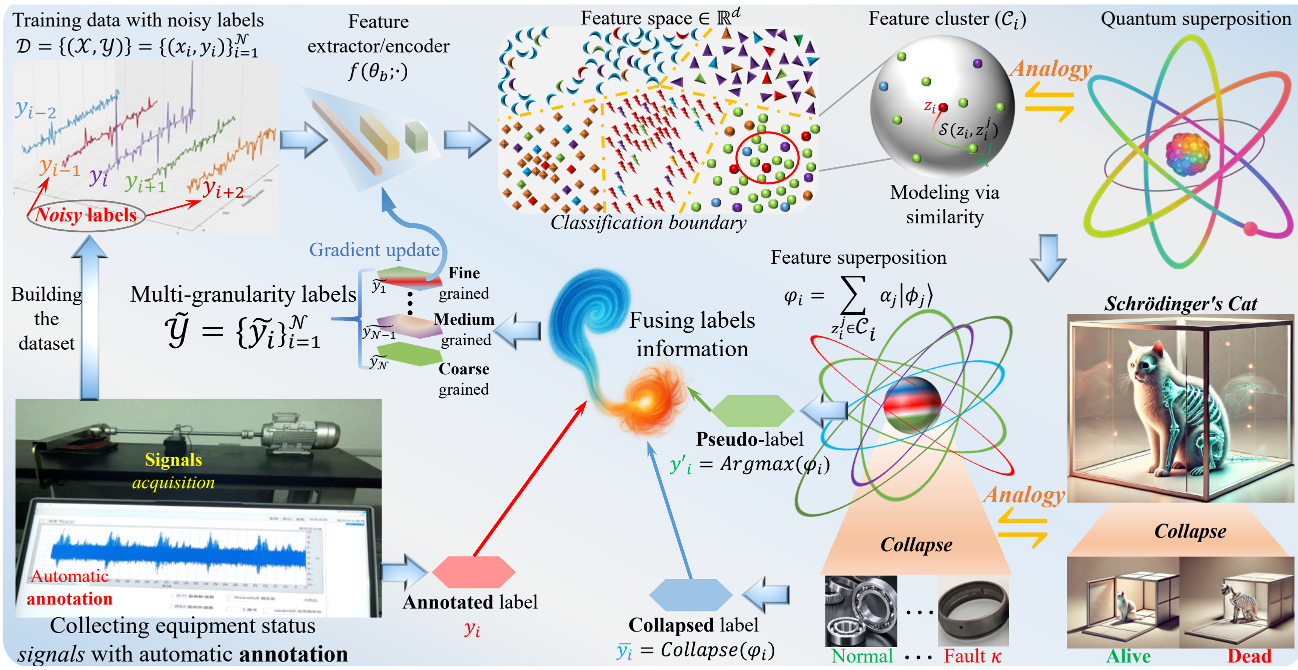


Fig. 2. The overview of MgL: MgL begins by mapping input signals to a latent feature space through the backbone of the diagnostic model. For each sample $\boldsymbol{x}_i$, a corresponding feature cluster $\boldsymbol{\mathcal{C}}_i$ is formed, and the feature distribution along with the label information within each cluster are combined to generate a superposition state φ_i, representing the cluster-center sample. Following the Copenhagen interpretation of quantum mechanics, the superposition state φ collapses into a specific eigenstate ϕ_j, resulting in a collapsed label. Meanwhile, a greedy algorithm is employed to select the eigenstate with the highest probability amplitude as the pseudo-label for the cluster-center sample. Finally, MgL re-labels the dataset by fusing the original annotated labels $\boldsymbol{\mathcal{Y}}$, collapsed labels $\overline{\boldsymbol{\mathcal{Y}}}$, and pseudo-labels $\boldsymbol{\mathcal{Y}}'$, and these newly generated multi-granular labels $\tilde{\boldsymbol{\mathcal{Y}}}$ are then used as supervision to train the diagnostic model.

noisy labels has become a key challenge to ensure robust performance in practical fault diagnosis scenarios [11].

In response to this challenge, **L**earning with **N**oisy **L**abels (*LNL*) methods have emerged to train deep neural networks effectively in noisy environments. These methods generally fall into three categories: robust loss function design [12], sample selection [13], and label correction [14]. Robust loss functions aim to mitigate the impact of noisy samples, enhancing training stability [15]. However, they often struggle when the noise ratio is high or when noise patterns are difficult to estimate, as they fail to differentiate clean from noisy samples effectively [13]. Due to these limitations, sample selection and label correction strategies have gained more traction and have been the focus of extensive research [11].

Most sample selection and label correction methods adopt a self-guided learning process. Initially, the entire dataset is used to warm up the model. Sample selection methods then iteratively identify clean samples (e.g., using small-loss criteria [16]) via the predictions of model, which are used for supervised learning in subsequent iterations [17]. In contrast, label correction methods refine the label information by combining observed labels with model predictions (such as latent features [18] or logits [19]) to dynamically optimize the dataset during training [20]. Despite their success, these strategies share a critical limitation [9]: they rely heavily on the model's predictions, which can accumulate confirmation bias during self-guided training cycles (Fig. 1).

Essentially, these models guide themselves based on predictions made in noisy environments. Incorrect predictions may propagate through successive training steps, amplifying bias and degrading performance. This limitation motivates the core question of this work:

*How can the **confirmation bias** caused by inaccurate predictions in LNL processes **be mitigated** to enhance the generalization capability of diagnosis models in high-noise environments?*

Inspired by the principles of quantum superposition and collapse, we propose a Multi-granularity Labeling (MgL, see Fig. 2) approach to mitigate bias in diagnosis models trained with noisy labels. In conventional classification tasks, deep learning models typically treat each sample as an independent "measurement", updating model parameters based on a single prediction pathway. However, when labels contain noise, this reliance on a single pathway often overlooks the essential multi-faceted information embedded in the feature space, leading to cumulative confirmation bias during training.

To address this issue, we conceptualize feature clusters $\mathcal{C}_i$ containing conflicting labels as a "superposition state"($\varphi = \sum_j c_j|\phi_j\rangle$), where each potential label weight α_j represents its "belief mass", analogous to the Born rule $P(j) = |c_j|^2 = \alpha_j$ governing measurement probabilities. During the generation of collapsed labels $\bar{y}_i$, MgL introduces controlled randomness to refine label assignment strategies, thereby reducing over-reliance on noisy labels. Additionally, MgL integrates stochastic collapse results, original annotations y_i, and pseudo-labels y'_i to construct multi-granularity labels $\tilde{y}_i$ (Fig. 3). This process mirrors repeated observations of a quantum system under different measurement bases (e.g., position and momentum), simulating multi-source information and reducing model sensitivity to individual erroneous predictions.

Notably, the entropy $\mathrm{H}(\tilde{y}_i)$ of multi-granularity labels dynamically adjusts with noise intensity η. When $\eta \rightarrow 0$, $\tilde{y}_i$ approximates a Dirac distribution ($\mathrm{H}(\tilde{y}_i) \approx 0$), corre-

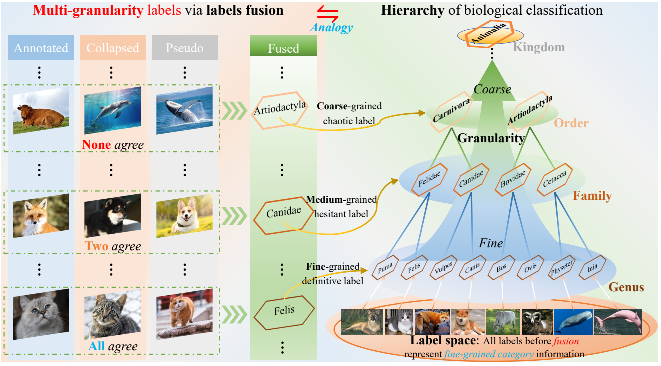


Fig. 3. An illustrative example of how MgL combines annotated, collapsed, and pseudo labels to form three granularity levels: When all three labels coincide (e.g., all identified as "cat"), a **fine**-grained definitive label (Genus level, *Felis*) is assigned; when exactly two labels coincide (e.g., "dog", "dog", and "fox"), a **medium**-grained hesitant label (Family level, *Canidae*) is produced; and when all three labels conflict (e.g., "bull", "dolphin", and "whale"), a **coarse**-grained chaotic label (Order level, *Artiodactyla*) is adopted.

sponding to a traditional deterministic decision mode (i.e., $\tilde{y}_i$ approximates a fine-grained definitive label). Conversely, as $\eta \to 1$, $\tilde{y}_i$ transitions into a high-entropy quantum-like random distribution ($\mathrm{H}(\tilde{y}_i)$ increases significantly), enhancing the model's exploratory capability in uncertain environments. This dynamic adaptation from deterministic to high-entropy superposition states is analogous to the quantum-classical transition, enabling MgL to adaptively regulate the "quantum nature" of label correction (maintaining high certainty in low-noise environments while promoting stochasticity in high-noise settings), which effectively mitigates confirmation bias and enhances the model's robustness.

Extensive experiments on proprietary and publicly available datasets validate the effectiveness and superiority of MgL. In summary, our contributions are threefold:

1) **Theory**: Inspired by quantum collapse, we introduce a mapping paradigm for converting feature cluster distributions into class information of sample, expanding the theoretical framework for information fusion in decision-making.
2) **Methodology**: We propose a multi-granularity labeling strategy that substantially improves the noise tolerance of diagnostic models in strong noise environments, offering a robust training paradigm for related fields.
3) **Engineering**: Comprehensive experiments demonstrate MgL's efficacy across various noise levels, highlighting its potential for broader applications in low-cost, automated diagnostic datasets.

II. Preliminary

A. Problem formulation (Fault diagnosis with noisy labels)

Fault diagnosis with noisy labels can be framed as a multi-class classification problem under label noise interference [5]. Specifically, it is assumed that the training dataset is contaminated with noise and consists of N industrial time-series samples, denoted as $\mathcal{D} = \{\mathcal{X}, \mathcal{Y}\} = \{(x_i, y_i)\}_{i=1}^{N}$. Each sample $x_i \in \mathbb{R}^{C\times L}$ represents a time-series signal with C channels and L consecutive sampling points, while $y_i \in [1, K]$ is the corresponding label. Let $\hat{y}_i$ represent the true class label for input x_i. If $\hat{y}_i \neq y_i$, the label y_i is considered noisy; otherwise, it is deemed clean. And the noise distribution $T(\mathcal{Y} \mid \hat{\mathcal{Y}})$ is typically unknown.

The goal of LNL in fault diagnosis is to design a strategy with noisy tolerance, to train a model $f(\theta_b, \theta_c; \cdot)$ — where θ_b denotes the backbone parameters and θ_c refers to the classifier parameters — such that the model performs well not only on the training data but also generalizes effectively to unseen data [21], which can be formalized by minimizing the difference between the diagnostic results $\mathcal{P} = f(\theta_b, \theta_c; \mathcal{X})$ and the true fault $\hat{\mathcal{Y}}$, expressed as $\underset{\theta_b,\theta_c}{\operatorname{argmin}} \left\{ \mathbb{E}\left(\mid \mathcal{P} - \hat{\mathcal{Y}} \mid \right) \right\}$.

B. Nomenclature

To ensure clarity and consistency throughout this article, this subsection defines the primary notations. Table I provides an overview of all major symbols.

TABLE I
KEY NOTATION AND DEFINITIONS

Symbol	**Definition**
$\mathcal{D} = \{\mathcal{X}, \mathcal{Y}\} = \{(x_i, y_i)\}_{i=1}^{N}$	Training dataset of N samples and their labels
$x_i \in \mathbb{R}^{C\times L}$	i-th time-series sample with C channels and L points
$y_i \in \{1, \ldots, K\}$	Original annotated (potentially noisy) label of x_i
$\hat{y}_i \in \{1, \ldots, K\}$	True (noise-free) label of x_i
$f(\theta_b; \cdot)$, $f(\theta_c; \cdot)$	Backbone and classifier, forming $f(\theta_b, \theta_c; \cdot)$
$z_i = f(\theta_b; x_i)$	Latent feature vector extracted by the backbone
$\mathcal{C}_i$	Feature cluster constructed around z_i based on similarity
ϕ_i	One-hot basis state derived from the observed label y_i
φ_i	Quantum superposition for label distribution in $\mathcal{C}_i$
$\bar{y}_i$	Collapsed label via wavefunction collapse on φ_i (Copenhagen)
y_i'	Pseudo label, i.e., the most probable category in φ_i
$\tilde{y}_i$	Multi-granularity label combining $\bar{y}_i$, y_i', and y_i

C. Related works

Deep neural networks (DNNs) readily overfit to noisy labels, causing severe performance deterioration in classification tasks—including fault diagnosis [22]. To address this, recent LNL research broadly focuses on designing robust loss functions, separating clean/noisy samples, and correcting noisy annotations [11].

1) Robust loss functions: primarily aim to lessen the impact of label noise on the gradient update process. For instance, NLS [23] employs label smoothing to reduce overconfidence in noisy annotations, SCE [15] adds a symmetric term to cross-entropy to balance noise influence, while AGCE [12] and RML [22] propose refined forms of asymmetric or regrouped loss for high-noise scenarios. However, these methods often presuppose an accurate or partially known noise structure (e.g., the transition matrix $T(\mathcal{Y} \mid \hat{\mathcal{Y}})$), which may be impractical to estimate in real industrial settings [24].

2) Sample separation: attempt to isolate relatively "clean" samples from noisy ones. Early approaches adopt small-loss selection—e.g., JoCoR [16] cross-updates two networks using mutually consistent low-loss instances—while DISC [25] filters out suspected noisy data based on memory strength. Fault diagnosis research has further introduced granular computing, as in MgBIF [6], to better handle uncertainty in label confidence. Yet, excessive removal of "noisy" samples can shrink the training set and lead to underfitting, which some methods (e.g., MgCNL [5]) counter through contrastive or unsupervised learning to salvage valuable information from the excluded data.

3) Label correction: utilize model predictions to refine noisy labels, thus alleviating the data scarcity issues in sample removal. Representative efforts, such as SED [24], combine mean-teacher frameworks with pseudo-labeling to iteratively update questionable annotations. Although effective in many high-noise cases, these algorithms can still accumulate prediction errors over repeated "self-corrections," exacerbating confirmation bias [9]. Recent studies, including ANNE [13], highlight that enhancing label refinement with regularization or adaptive neighbor consistency can reduce such drift.

Despite these advances, achieving robust label correction for fault diagnosis under intense noise remains challenging. To this end, our proposed MgL introduces a quantum-collapse-inspired multi-granularity strategy that maximally exploits original labels—even if noisy—alongside pseudo-labels and stochastically "collapsed" labels. By embedding inherent uncertainty into the correction pipeline, previous erroneous predictions can be partly overridden in subsequent updates, curbing confirmation bias (see Fig. 1) and boosting high-noise robustness.

D. Copenhagen interpretation and Multi-granularity labeling

In the Copenhagen interpretation of quantum mechanics [26], a quantum system is described by a superposition of eigenstates before measurement:

$$\varphi = |\psi\rangle \;=\; \sum_{k=1}^{K} c_k \;|\phi_k\rangle \tag{1}$$

where $\{|\phi_k\rangle\}_{k=1}^{K}$ denotes the set of orthonormal eigenstates, and c_k are complex probability amplitudes. The Born rule states that the squared magnitude of each amplitude, $\alpha_k = |c_k|^2$, gives the probability of the system collapsing to the corresponding eigenstate $|\phi_k\rangle$ upon measurement. Thus, while the system initially remains in a "superposed" configuration, a single measurement forces it to collapse into a definite state, and this outcome is intrinsically probabilistic.

Drawing inspiration from this collapse process, MgL analogizes the label assignment mechanism to quantum measurement. Specifically, we treat the label distribution of each feature cluster $\mathcal{C}_i$ as a superposition state of the cluster-center sample x_i. Suppose that $\mathcal{C}_i$ contains $|\mathcal{C}_i|$ samples with corresponding labels $\{y_j\}_{j=1}^{|\mathcal{C}_i|}$. We can define a set of label "eigenstates" $\{|\phi_k\rangle\}_{k=1}^{K}$ corresponding to the K possible classes. Inspired by the Born rule, we introduce a probability distribution for cluster $\mathcal{C}_i$:

$$\alpha_k(\mathcal{C}_i) \;=\; \frac{1}{|\mathcal{C}_i|} \sum_{x_j \in \mathcal{C}_i} \mathbb{1}(y_j = k) \tag{2}$$

where $\mathbb{1}(\cdot)$ is an indicator function that equals 1 if the condition holds and 0 otherwise, and $\alpha_k(\mathcal{C}_i)$ represents the "belief mass" that the true label of x_i is class k. MgL then performs a *collapse* by sampling a specific class $\bar{y}_i$ from this distribution:

$$\bar{y}_i \;\sim\; \mathrm{Cat}\Big(\alpha_1(\mathcal{C}_i), \ldots, \alpha_K(\mathcal{C}_i)\Big), \tag{3}$$

which parallels the notion of choosing one eigenstate with probability $\alpha_k(\mathcal{C}_i)$. By doing so, the label of x_i is not deterministically assigned to the most frequent class in $\mathcal{C}_i$; instead, it is stochastically "observed" from the cluster's distribution. This modeling choice brings beneficial randomness into the label assignment, preventing the model from fixating on potentially dominant but noisy label patterns.

In classical fault diagnosis pipelines, one risks reinforcing erroneous annotations when labels are taken to be unequivocally correct—especially under high noise levels. By contrast, MgL's controlled stochastic selection (Eq. (3)) ensures that label fusion does not deterministically converge to a single, potentially incorrect class; instead, multiple plausible classes receive due consideration. This stochasticity mitigates the self-reinforcement of mislabeling—akin to reducing confirmation bias—while still guiding the cluster toward a more consistent labeling scheme.

Through this lens, MgL may be regarded as an instantiation of "*Certainty from Uncertainty*", wherein short-term randomness in the collapsed label assignment ultimately strengthens the system's overall determinacy. This is achieved by allowing the model to discover robust label correlations rather than relying on a single, possibly flawed vote, thereby improving performance and stability in noisy-label scenarios.

III. Methodology

A. Design rationale of MgL

According to Refs [5], [6], [9], DNNs with strong generalization capabilities tend to form distinct feature clusters in their latent feature space, and these clusters can be easily separated by category. Intuitively, if two samples are very close in the feature space, their classification results should also be similar, implying that their true class information is likely highly consistent (Assumption 1).

Assumption 1: (**Label consistency**) *Let $Sim(z_i, z_j)$ represent the similarity between z_i and z_j. For $\forall x_i, x_j \in \mathcal{D}$, if $f(\theta_b, \theta_c; \cdot)$ is sufficiently smooth and generalizes well, then:*

$$\begin{gathered} Sim(z_i, z_j) \geq Sim(z_i, z_i) - \varepsilon = Sim(z_j, z_j) - \varepsilon \\ \Downarrow \\ P(\hat{y}_i = \hat{y}_j) \approx P(f(\theta_b, \theta_c; x_i) = f(\theta_b, \theta_c; x_j)) \geq 1 - \tau \cdot \varepsilon \end{gathered} \tag{4}$$

where $\varepsilon \geq 0$ represents an infinitesimal value, while $\tau > 0$ is a classifier-dependent constant. A smaller τ indicates a greater distributional divergence between features of different classes. Additionally, $P(\hat{y}_i = \hat{y}_j)$ denotes the probability of true class consistency.

Based on this, for any sample x_i with annotated label y_i to be corrected, the MgL first constructs a feature cluster $\mathcal{C}_i$ in the tent feature space centered on z_i using a similarity-based approach. The label $\mathcal{Y}_i$ of other samples within $\mathcal{C}_i$ are treated as the ground states into which the superposition state of sample x_i collapses upon measurement. By integrating the feature distribution $\mathcal{Z}_i$ with $\mathcal{Y}_i$ within the cluster, MgL characterizes x_i in a superposition state φ_i and selects the category with the highest amplitude probability as the pseudo-label (when most samples in $\mathcal{C}_i$ belong to a certain category, x_i is also likely to belong to that category).

However, during training, diagnostic model cannot perfectly predict the true labels of all samples, and classifier smoothness can fluctuate over time. I.e., directly replacing the original annotated labels with pseudo-labels generated from the superposition state may introduce confirmation bias, especially in noisy environments, thereby impairing model generalization. To address this issue, MgL observed the superposition again to generate collapsed label $\overline{y_i}$ with uncertainty and referred to some of the annotated label, to weaken the belief of pseudo label, thus constructing a three-granularity labeling framework to optimize model training.

Specifically, MgL defines three types of labels (Fig. 3) based on the consistency between pseudo-labels, collapsed labels, and annotated labels and this three-granularity framework reduces biases introduced by pseudo-labels while leveraging redundant label information, enhancing the robustness and generalization of the diagnostic model.

B. Constructing feature clusters

MgL considers each non-center sample $z_j \in \mathcal{C}_i$ ($z_i \neq z_j$) within a feature cluster $\mathcal{C}_i$ as the result of observing the cluster center z_i under specific conditions. Due to the high similarity between z_i and z_j in the feature space, z_j can approximately substitute z_i (i.e., $z_i \approx z_j$). The corresponding label y_j is treated as the observed collapsed state. Based on this principle, MgL constructs a feature cluster $\mathcal{C}_i$ for each sample x_i in the latent feature space ($z_i = f(\theta_b; x_i)$), simulating the results of multiple observations of the sample under parallel conditions.

To facilitate matrix-based acceleration of similarity computations and to amplify the differences among highly similar samples, MgL defines an exponential-scaled similarity function based on cosine similarity (Eq. (5)), which ensures that the similarity between samples of the same class is significantly higher than that of samples from different classes, even when the cluster center sample is located near the edge of the class distribution:

$$Sim(z_i, z_j) = e^{(\cos(\langle z_i, z_j \rangle) + \widetilde{\eta})} = e^{\widetilde{\eta}} \cdot e^{\left(\frac{z_i \times z_j^\top}{\|z_i\| \cdot \|z_j\|}\right)} \tag{5}$$

For clarity, we introduce a new notation: let $z_i^{(n)}$ denote the n-th most similar feature to z_i, i.e., $\forall z_i \in \mathcal{Z}, n \in \mathbb{N}^+, s.t. Sim(z_i, z_i^{(n)}) > Sim(z_i, z_i^{(n+1)})$. Based on this, the cluster $\mathcal{C}_i$ centered on z_i is defined as:

$$\begin{cases} \mathcal{C}_i & = \{z_j \mid Sim(z_i, z_j) \geq Sim(z_i, z_i^{(n)})\} \\ n & = \max(\text{Initialize}(n) \times \widetilde{\eta}, 10) \end{cases} \tag{6}$$

where n determines the number of features included in $\mathcal{C}_i$ (in this work, Initialize$(n) = 2^7$), and $\widetilde{\eta} = \frac{1}{N}\sum_{i=1}^{N} Max(\varphi_i)$ represents the estimated noise intensity of the dataset. The constant value 10 serves as a truncation threshold to prevent the extreme case where n approaches zero when the noise intensity $\widetilde{\eta}$ is too low, which ensures that the feature cluster $\mathcal{C}_i$ retains a sufficient diversity of label information.

The value of $\widetilde{\eta}$ is dynamically adjusted during training via predictions from the previous epoch (see Sec. III-D), to better align with the requirements of feature cluster construction: noisier datasets require larger clusters to comprehensively describe the state of the cluster center, while less noisy datasets allow for smaller clusters to reduce computational overhead.

C. Quantum state tomography via cluster fusion

In quantum mechanics, Quantum State Tomography [27] reconstructs quantum state information, including the amplitude probabilities of basis states in a superposition, by sampling the probability distributions under different measurement bases. Inspired by this process, MgL treats the labels corresponding to features within a feature cluster as the collapsed basis state information obtained from individual measurements. By integrating information from all samples in a cluster, MgL constructs a superposition state to represent the probabilities of a sample collapsing to each category.

To enhance the robustness and accuracy of the superposition state, MgL leverages the concept of evidence combination from the Dezert-Smarandache theory [28]. Each sample within a cluster is treated as an independent source of evidence, and a similarity-based intra-cluster fusion process is designed, as described by the following:

$$\begin{cases} \varphi_i & = \sum_{k=1}^{K} \alpha_k \cdot |\phi_k\rangle, \\ |\phi_k\rangle & = \text{One-hot}(k),\ k \in [1, K], \\ \alpha_k & = \frac{\sum_{z_j \in \mathcal{C}_i} Sim(z_i, z_j) \cdot \mathbb{1}(y_j = k)}{\sum_{z_j \in \mathcal{C}_i} Sim(z_i, z_j)}, \end{cases} \tag{7}$$

where K represents the total number of categories.

In other words, for $\forall x_i \in \mathcal{D}$, after constructing its feature cluster $\mathcal{C}_i$, MgL uses the similarity $Sim(z_i, z_j)$ as the belief mass (also call fusion weight) for the annotated label y_j ($z_j \in \mathcal{C}_i$), combining the labels and feature distributions of other samples within the cluster to determine the amplitude probabilities of x_i collapsing to each basis state $|\phi_k\rangle$ (i.e., this superposition state φ_i represents the probabilities of the sample belonging to different categories).

By adopting this similarity-based fusion method, MgL incorporates both intra-cluster feature distributions and label frequency information during the reconstruction of the superposition state. This results in more accurate and robust category determination, even in noisy and uncertain environments.

D. Multi-granularity labels

After obtaining the superposition state φ_i, MgL applies the maximum a posteriori decision rule commonly used in multi-class deep learning tasks to generate the pseudo-label y_i' and records the amplitude probability $a_i = \alpha_{y_i'}$ of the corresponding basis state (see Eq. (8)). This amplitude probability approximates the frequency of the major category within the feature cluster and is referred to as intra-cluster label consistency. MgL, therefore, estimates the noise intensity η of the dataset $\mathcal{D}$ by computing the global mean of a_i, i.e. $\widetilde{\eta} = \text{Mean}(a_i)$ in Eq. (5) and Eq. (6):

$$y_i' = \arg\max(\varphi_i), \quad a_i = \max(\varphi_i) \tag{8}$$

Similarly, the amplitude α_{y_i} of the basis state $|\phi_{y_i}\rangle$ corresponding to the annotated label y_i in φ_i is defined as the certainty of the annotated label, $c_i = \alpha_{y_i}$.

To suppress confirmation bias, which may arise from directly using pseudo-labels to correct the original annotated labels, MgL simulates a random collapse process by resampling the superposition state φ_i, which generates a stochastic collapsed label $\overline{y_i}$, where the sampling probabilities for each class are consistent with the amplitude probabilities of the corresponding basis states in φ_i. Unlike conventional methods that rely on deterministic voting or fixed probability distributions, MgL leverages the principles of quantum superposition and collapse to mitigate confirmation bias and enhance model generalization. Specifically, each sample's label is treated as a probabilistic outcome influenced by its neighborhood, where surrounding samples serve as potential alternative "measurement results" . During training, MgL first constructs a superposition state by encoding the label distribution within the sample's feature cluster, representing the likelihood of each class. Subsequently, the collapse process is employed, preventing the model from being strictly confined to the most frequent class prediction, which allows MgL to counteract cumulative annotation bias over successive training iterations. By integrating collapsed labels into multi-granularity supervision, MgL effectively reduces the self-reinforcement of erroneous predictions, thereby enhancing robustness in high-noise environments.

Next, MgL fuses the pseudo-label y_i', collapsed label $\overline{y_i}$, and annotated label y_i to construct the final label $\widetilde{y_i}$ via Eq. (9). The fusion weights for the pseudo-label and annotated label are directly determined by their amplitude probabilities a_i and c_i, respectively, while the weight of the collapsed label is adjusted based on its relative credibility as a simulated random observation.

$$\widetilde{y_i} = \begin{cases} (1-a_i-c_i)\cdot\overline{y_i} + a_i\cdot y_i' + c_i\cdot y_i, & \text{if } y_i' \neq y_i, \\ (1-a_i)\cdot\overline{y_i} + a_i\cdot y_i', & \text{otherwise.} \end{cases} \tag{9}$$

Note: $\overline{y_i}$, y_i', and y_i must be one-hot encoded before fusion.

To summarize, MgL seeks to create a more robust labels, $\widetilde{\mathcal{Y}}$, via combining the statistical consistency of pseudo-labels, the randomness introduced by collapsed labels, and the original information from annotated labels, which mitigates confirmation bias typically associated with label correction based on predictions, thereby enhancing the model's generalization ability, especially in high-noise environments.

E. Overview

MgL (Alg. 1) aligns with mainstream label correction strategies in LNL and follows a standard supervised learning process during training.

The key difference lies in the pre-epoch phase, where MgL simulates a Quantum State Tomography process based on the distribution of the current model in the latent feature space, which generates superposition states to describe the class attributes of all samples. Then using these superposition states, MgL constructs multi-granularity labels to replace the original ones, facilitating the subsequent gradient update process.

Algorithm 1 Training process based on MgL

Input: Number of $Epochs$, the training dataset ($\mathcal{D} = \{\mathcal{X}, \mathcal{Y}\}$), estimating noise intensity ($\widetilde{\eta}$), and the diagnostic model ($f(\theta_b, \theta_c; \cdot)$).
Output: $f(\theta_b, \theta_c; \cdot)$ (Trained diagnostic model)

Random initialization diagnostic model $f(\theta_b, \theta_c; \cdot)$, and initialization estimating noise intensity ($\widetilde{\eta} = 0.9$).
for $epoch$ in range(Total $Epochs$) **do**
  $\mathcal{Z} = f(\theta_b; \mathcal{X})$ *# Get latent features*
  $\mathcal{C}s = \{\mathcal{C}_i\}_{i=1}^{N} \leftarrow \widetilde{\eta}, \mathcal{Z}$ via Eq. (6) *# Get feature clusters*
  $\varphi s = \{\varphi_i\}_{i=1}^{N} \leftarrow \mathcal{C}s$ via Eq. (7) *# Quantum state tomography*
  Update $\widetilde{\eta}$ via φs: $\widetilde{\eta} = \underset{\varphi_i \in \varphi s}{\text{Mean}}[Max(\varphi_i)]$
  $\overline{\mathcal{Y}}$ = *one_hot*(*multinomial*(φs)) *# Get collapsed labels*
  $\mathcal{Y}', A$ = *max*(φs, dim=1) *# Get pseudo-labels with amplitude*
  $\mathcal{Y}'$ = *one_hot*($\mathcal{Y}'$), $\mathcal{Y}$ = *one_hot*($\mathcal{Y}$) *# Update format of labels*
  $C = (\varphi s \cdot \mathcal{Y})$.*sum*(dim=1) *# Get certainty of annotated labels*
  $chaos = 1 - (\varphi s \cdot (\mathcal{Y}' + \mathcal{Y})$.*clip*(0, 1)).*sum*(dim=1)
  $\widetilde{\mathcal{Y}} = A \times \mathcal{Y}' + C \times \mathcal{Y} + chaos \times \overline{\mathcal{Y}}$ via Eq. (9)
  # Training diagnostic model via ***Multi-granularity labels*** $\widetilde{\mathcal{Y}}$
  $\widetilde{\mathcal{D}} = \{\mathcal{X}, \widetilde{\mathcal{Y}}\}$ *# Replace the original labels*
  for $(x_i, \widetilde{y_i})$ in $\widetilde{\mathcal{D}}$ **do**
    $p_i \leftarrow f(\theta_b, \theta_c; x_i)$
    $loss \leftarrow \mathcal{L}_{CE}(p_i, \widetilde{y_i})$
    Update θ_b, θ_c via $loss$
  end for
end for

IV. Experimental verification and discussion

A. Experimental settings

1) Dataset description: To comprehensively evaluate the proposed method, we conducted experiments on five datasets: *CWRU* [29], *XJTU* [30], *NEPU* [31], *Multiple* [32], and *Custom*. The first four are open-source datasets, while Custom is a self-collected dataset designed to augment the diversity and complexity of the experimental setup. All datasets consist of vibration signals directly collected from mechanical systems during operation, covering multiple working conditions under varying rotational speeds and load conditions.

The datasets were selected to represent a wide range of fault scenarios and complexities. The details of each dataset, including fault categories, working conditions, and sample sizes, are summarized in Table II. For this work, we intentionally avoided addressing the issue of data imbalance; thus, the number of training and testing samples for each category is approximately equal. Each sample contains 2048 continuous data points, and there is no overlap between any two samples, ensuring a clean and consistent evaluation framework.

TABLE II
THE DETAILED INFORMATION ABOUT ALL DATASETS

Parameter	Detailed information about *Custom* dataset
Motor speed	[1770, 1775] rpm, [2366, 2370] rpm, [2959, 2962] rpm
Motor load	0.0 NM, and 0.3 NM
Fault location	Inner, Outer, Ball, Inner and Outer, Inner and Ball, Outer and Ball
Damage degree	0.2 mm, 0.4 mm, 0.6 mm, and 0.8 mm
Sampling frequency	16 kHz
No. of categories	$1 + (3 + 3) \times 4 = 25$
No. of samples for train	$25 \times 6 \times 376 = 56,400$
No. of samples for validation	$25 \times 6 \times 187 = 28,050$
No. of samples for test	$25 \times 6 \times 376 = 56,400$

Parameter	Detailed information about open-source datasets			
	CWRU	*XJTU*	*NEPU*	*Multiple*
No. of categories (faults+1)	12	9	7	49
No. of working condition	4	2	4	6
Sampling frequency	48 kHz	25.6 kHz	12 kHz	12 kHz
No. of tra. samples	10,885	8,252	3,584	76,734
No. of val. samples	5,439	4,116	1,820	25,578
No. of tes. samples	10,885	8,252	3,584	76,734

To assess the robustness of MgL against noisy labels, two types of noise were introduced: symmetric noise (class-independent) and asymmetric noise (class-dependent). Symmetric noise was simulated using random label replacement at five noise levels (η = 50%, 60%, 70%, 80%, and 90%), while asymmetric noise was generated by swapping labels between similar classes at five levels (η = 25%, 30%, 35%, 40%, and 45%). To simulate asymmetric noise, labels were swapped between similar classes to mimic realistic scenarios where mislabeling is more likely to occur. For example, in the Multiple dataset, the fault location labels were swapped between proximal and distal bearings while keeping the fault type unchanged. Similarly, in the Custom dataset, the damage severity labels for the same fault category were interchanged, such as swapping 0.2 mm with 0.4 mm. These noise settings were designed to mimic realistic scenarios where label noise can significantly impact model performance.

2) Implementation details: As shown in Table III, to facilitate comparison, we utilized MgNet [32], an open-source fault diagnosis network released alongside the Multiple dataset, as the backbone to evaluate the performance of MgL.

TABLE III
THE IMPLEMENTATION DETAILS

Config	Value
Backbone	MgNet [32]
Optimizer	AdamW
Initial learning rate	$3 \times 10^{-3} \times \frac{batchsize}{512}$
Weight decay	10^{-2}
Optimizer momentum	$\beta_1 = 0.9$, $\beta_2 = 0.999$
Learning rate schedule	CosineAnnealingLR
Warm-up epochs	5
Training epochs	100
GPU	NVIDIA A100
Pytorch version	2.1.1+cu118

Prior to formal training, all models underwent a warm-up phase of 5 epochs, during which only standard supervised learning procedures were employed, without any additional operations for LNL. To prevent misleading results due to random initialization, all experimental outcomes reported in this section are based on statistical data from at least ten repeated trials, and the detailed experimental results are presented in the following sections.

B. Case I: Effectiveness of MgL

To demonstrate the effectiveness and superiority of MgL, we evaluated its performance across ten different noise environments on five datasets (Table II) and compared it with seven representative advanced LNL strategies (see Fig. 4), including SCE (2019 [15]), JoCoR (2020 [16]), RRL (2021 [33]), NCR (2022 [34]), AGCE (2023 [12]), RML (2024 [22]), and ANNE (2025 [13]).

As illustrated in Fig. 4, MgL consistently outperforms the baseline and most competing methods across various noise levels. Specifically, MgL achieves superior accuracy in high-noise environments ($\eta > 70\%$), significantly surpassing other methods. For example, on *Custom* with $\eta = 90\%$, MgL achieves an accuracy of 89.83$\pm$2.68%, while the best-performing competitor, ANNE, achieves 65.07$\pm$1.84%,. Similarly, on *Multiple* with $\eta = 90\%$, MgL achieves 56.46$\pm$4.86%, compared to ANNE's 29.64$\pm$1.86%,.

In low-noise environments, MgL remains competitive. On *CWRU* with $\eta = 25\%$, MgL achieves 99.62$\pm$0.29%, slightly surpassing ANNE's 99.56$\pm$0.38%. Notably, MgL's performance is robust even under severe label noise, demonstrating its effectiveness in mitigating the impact of confirmation bias in LNL methods. This robustness is particularly evident in high-noise scenarios, where MgL's multi-granularity labeling strategy significantly reduces performance degradation.

These results not only demonstrate the robustness and effectiveness of MgL, particularly in challenging high-noise environments, but also indicate that our collapse-based multi-granularity labeling strategy effectively mitigates the performance degradation caused by confirmation bias in LNL methods under high-noise conditions.

C. Case II: Ablation for MgL

To further examine the sources of MgL's effectiveness, we conducted ablation experiments by selectively removing the sparse factor, pseudo-label, and collapsed label from MgL. The results of these experiments, averaged over five datasets, are summarized in Table IV and detailed in Fig. 5. Here, removing the *sparse factor* means reverting Eq. (5) to standard cosine similarity.

TABLE IV
THE RESULTS OF THE ABLATION ON MGL

Method	Sparse factor	Pseudo label	Collapsed label	Average accuracy(%)
Baseline				70.43
A1		√	√	85.54
A2	√		√	81.60
A3	√	√		82.77
A4		√		81.35
A5			√	79.84
MgL	√	√	√	88.73

As shown in Fig. 5, each of the three components plays a crucial role in boosting MgL's performance under label noise. Removing any single component causes a discernible drop in accuracy, and removing multiple components compounds this decline. Specifically, the sparse factor helps adjust how strongly different features in the cluster contribute when forming the superposition state, thereby refining the confidence assigned to each potential label. Although removing it (A4 or A5) does not always inflict as large a drop as removing the pseudo-label, its absence still undermines MgL's resilience, especially when combined with the removal of other modules (e.g., comparing MgL $vs.$ A1: 88.73% $vs.$ 85.54%, A2 $vs.$ A5: 81.60% $vs.$ 79.84%, and A3 $vs.$ A4: 82.77% $vs.$ 81.35%).

The pseudo-label proves vital for correcting labels based on the most probable class memberships inferred from the superposition state. This correction is particularly beneficial under lower noise levels, where high-accuracy label refinement amplifies MgL's overall performance. Consequently, experiments that eliminate the pseudo-label (A2, A5) show more pronounced performance drops in lower noise conditions, reinforcing that pseudo-labeling is a key contributor to MgL's effectiveness in such scenarios.

Lastly, the collapsed label acts as MgL's primary safeguard against confirmation bias. By infusing controlled uncertainty into the learning process, it helps mitigate the risk of overfitting to noisy or partially incorrect labels. While this mechanism offers a crucial advantage in high-noise settings, its

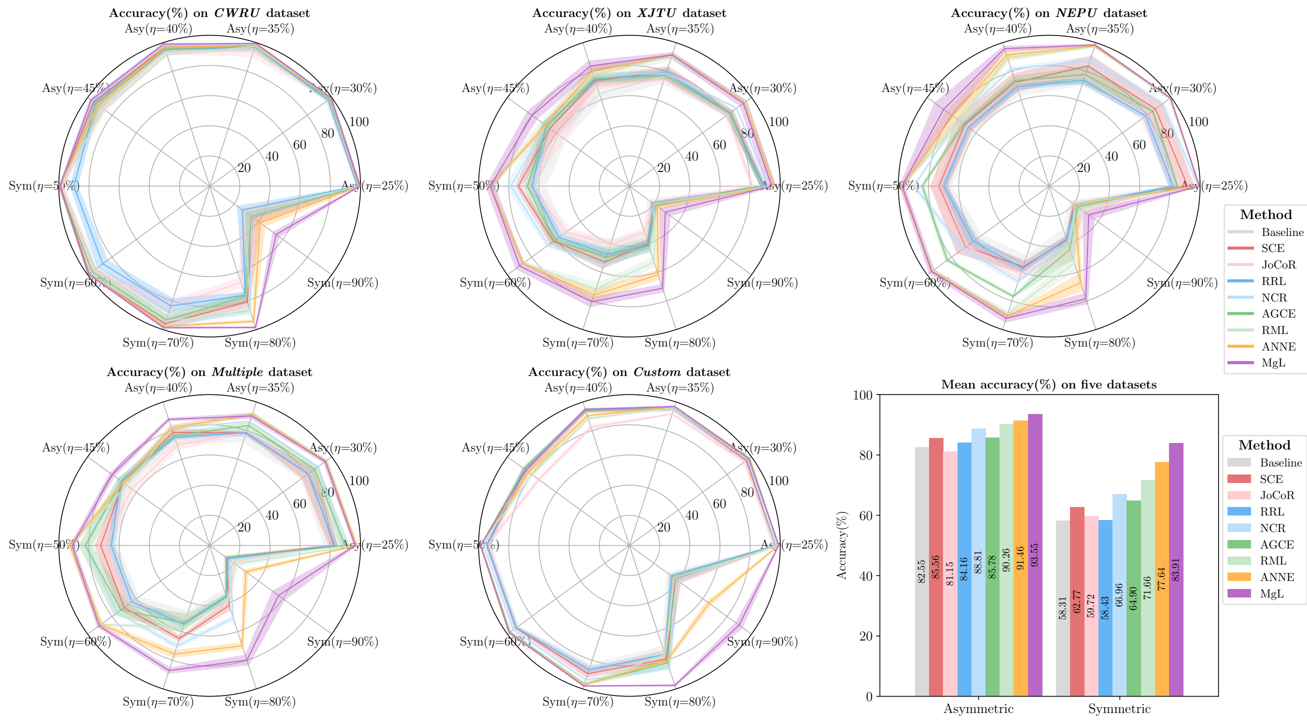


Fig. 4. Comparison of the effectiveness of MgL and other methods in a series of label noise environments. Where the light shaded area represents the accuracy fluctuation range of the corresponding method, and baseline represents the performance achieved via supervised learning without any LNL strategy.

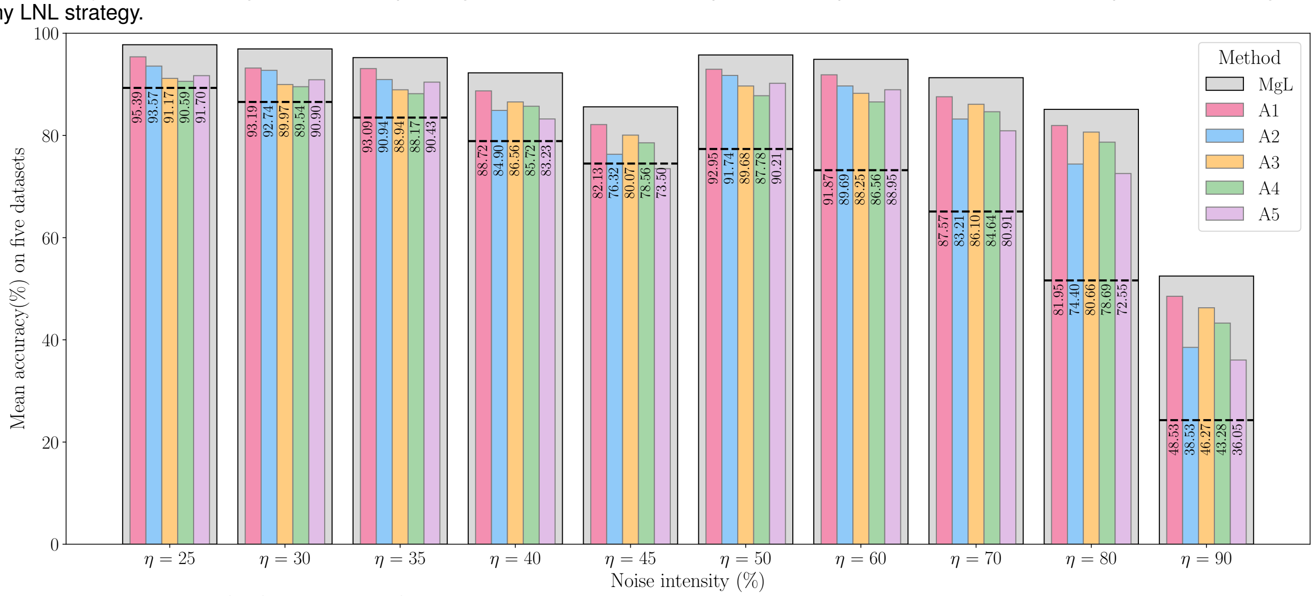


Fig. 5. Ablation results for fusing sparse factor, pseudo label, and collapsed label in generating multi-granularity labels, where dashed lines represent the accuracy of baseline.

standalone usage in lower noise environments may introduce superfluous uncertainty if not paired with pseudo-labels. The combined inclusion of both pseudo-labels and collapsed labels, as in full MgL, consistently yields the highest accuracy. This synergy underscores the necessity of jointly quantifying and fusing different label confidences for robust performance.

D. Case III: Sensitivity of hyperparameters

To investigate the sensitivity of MgL to key hyperparameters, we fixed the similarity scaling factor $\widetilde{\eta}$ (see Eq. (5)) and the sample size n in each feature cluster (see Eq. (6)) to several discrete values and measured the model's average accuracy under various noise levels. As illustrated in Fig. 6, MgL maintains reasonable robustness to both $\widetilde{\eta}$ and n across most noise scenarios, but its accuracy degrades significantly at extreme parameter values (e.g., $\widetilde{\eta}$ approaching positive or negative infinity, or an excessively large or small n). In contrast, the dynamic updating strategy—labeled "Dynamic" in the figure—consistently achieves superior performance under all tested noise conditions.

A closer examination reveals that, when $\widetilde{\eta}$ grows excessively large, samples near the cluster center acquire a disproportionate influence on cluster-level labels, effectively reducing the pseudo-labels and collapsed labels to mere replications of the center sample's category. This neglects valuable edge information in the feature space, undermining the model's ability to capture the true class distribution. Conversely, making $\widetilde{\eta}$ strongly negative causes the similarity to behave like a basic

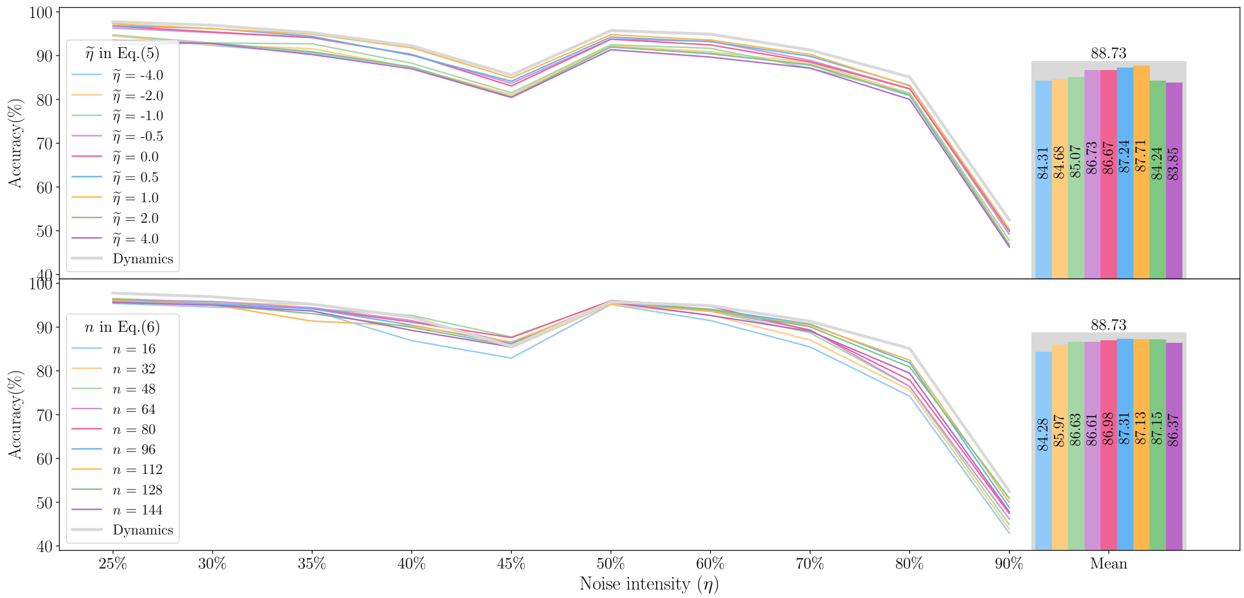


Fig. 6. Sensitivity of MgL to similarity scaling factor ($\widetilde{\eta}$ in Eq. (5)) and sample size in feature clusters (n in Eq. (6)), where "Dynamic" represents the current setting of MgL and "Accuracy" is the average accuracy on the five datasets.

voting mechanism. While counting alone does capture part of the label distribution, it discards the informative notion of "distance", leading to a decrease in overall accuracy.

Likewise, choosing an overly small n limits the cluster's information, leaving the pseudo-labels and superposition states highly vulnerable to a few misannotated samples. On the other hand, a very large n can cause clusters to overextend, incorporating samples from different class boundaries; in extreme cases, nearly all samples may merge into a single cluster, thus compromising both the collapsed and pseudo-labels.

E. Case IV: Visualization Analysis

This subsection evaluates the proposed Assumption 1 and the effectiveness of MgL under noisy conditions from two perspectives. First, we examine how feature similarity relates to label consistency across all test samples for the Baseline model trained with various noise levels (η), as shown in Fig. 7. Second, under a high noise ratio ($\eta = 90\%$), we select two datasets (Custom and XJTU) that exhibit the most significant performance gap and use t-SNE to visualize the latent feature distributions of Baseline and MgL (Fig. 8).

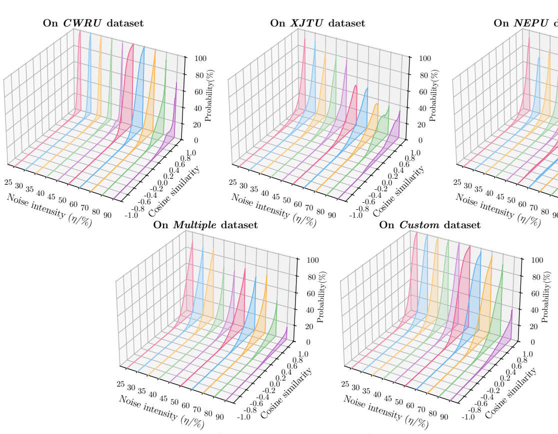


Fig. 7. In the latent feature space of the baseline model, the cosine similarity between any two features is positively correlated with the probability of their labels being consistent: the higher the similarity, the greater the likelihood of label consistency.

Fig. 7 demonstrates a positive correlation between cosine similarity $\cos(\langle z_i, z_j \rangle)$ and the probability that $\hat{y}_i = \hat{y}_j$ at all noise levels, consistent with Eq. (4), which indicates that, when the feature extractor $f(\theta_b; \cdot)$ has sufficient smoothness and strong generalization capability, samples with similar features are more likely to share the same true label. However, as η increases, the classifier's separation capability τ also rises, which causes a marked drop in label consistency among highly similar samples, which weakens the premise of Assumption 1 and explains why Baseline fails in high-noise scenarios.

To further investigate this issue and highlight MgL's advantages, we compare the t-SNE visualizations of Baseline and MgL on the Custom ($K = 25$) and XJTU ($K = 9$) datasets, both with $\eta = 90\%$ (Fig. 8). MgL significantly sharpens class boundaries in both datasets, confirming its theoretical soundness and practical value.

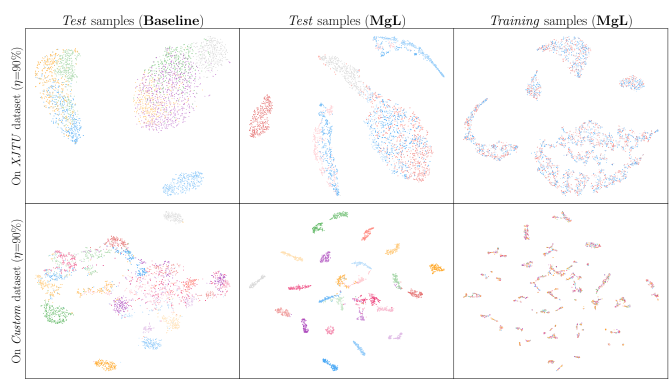


Fig. 8. The visualization of the latent feature space of the trained via t-SNE.

In particular: 1) When K is small and τ is high, Baseline on XJTU experiences severe class overlap, indicating that the principle "high similarity $\implies$ same class" is easily disrupted by strong noise; 2) On Custom, Baseline preserves a comparatively better inter-class distribution because noise is randomly dispersed across more categories, diluting the impact on individual classes; 3) In both datasets, MgL markedly enhances inter-class separability. By incorporating the uncer-

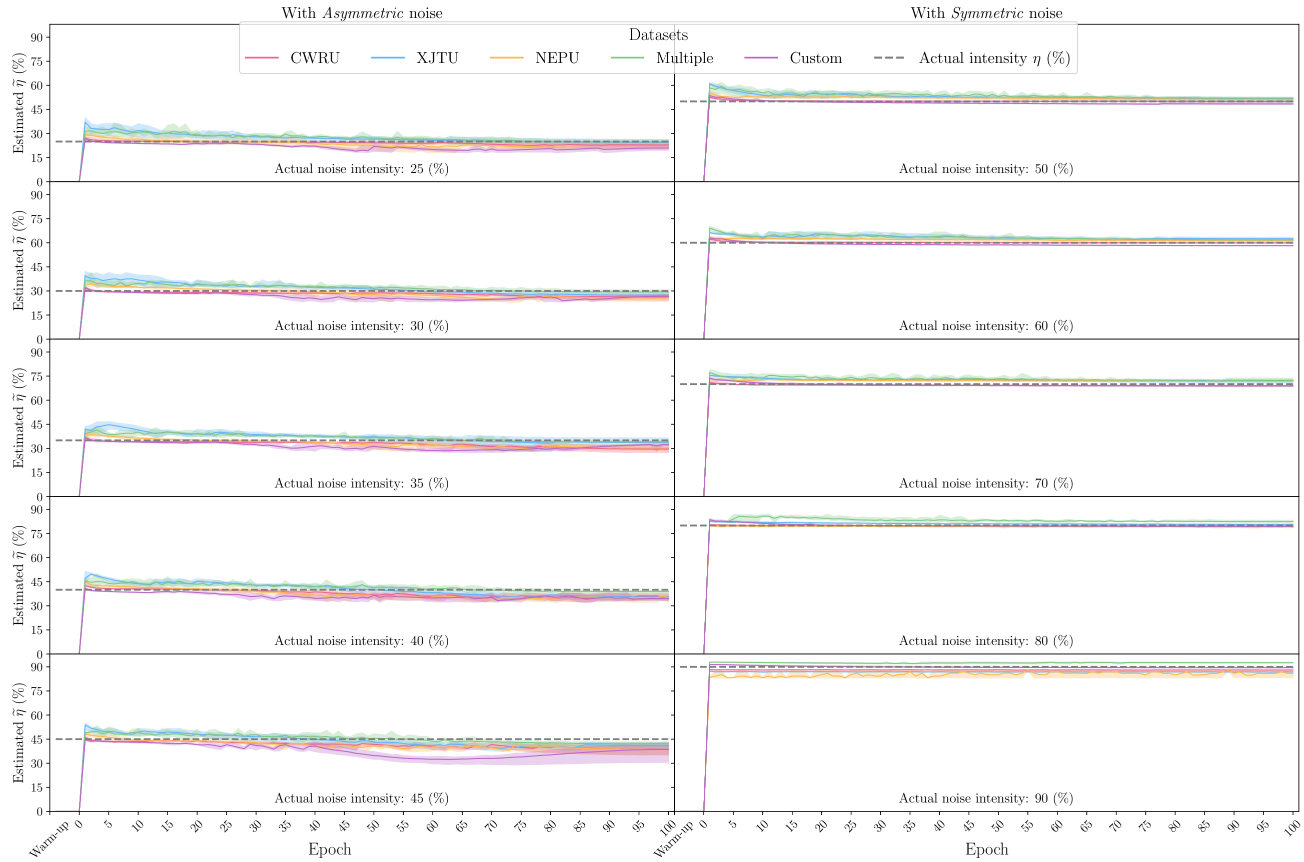


Fig. 9. The estimated noise intensity ($\widetilde{\eta}$) by MgL during the training process.

tainty of quantum collapse and integrating multi-granularity information, it reduces reliance on an absolute majority of correct labels and further relaxes the theoretical lower bound of "correct label dominance" $\big(\eta < (K-1)/K\big)$ to $P(\hat{y}_i = \hat{y}_j) \approx 1 - \varepsilon \cdot \tau > \frac{\varepsilon \cdot \tau}{K-1}$. Thus, even without an overwhelming majority of correct labels, the model still maintains high inter-class separability in feature space.

Fig. 9 plots the estimated noise intensity $\widetilde{\eta}$ versus training epochs for different true noise ratios (η). Although early training epochs (e.g., epoch < 10) sometimes exhibit a 6–10% deviation—due to still-evolving feature clusters—our estimation error for $\widetilde{\eta}$ consistently falls within 3.5% of the ground truth for the vast majority of epochs. This demonstrates both the overall effectiveness of our update strategy and its relatively strong robustness, even when clusters are not yet fully stabilized.

F. Limitations analysis

Despite MgL's effectiveness, several limitations warrant attention.

1) Computational overhead: Constructing superposition states φs relies on global feature-space statistics, requiring feature clustering for each sample, which can incur high computational and memory costs, particularly on large-scale or streaming datasets. Approximate nearest-neighbor search or localized partitioning may be needed to enhance scalability in future work.

2) Focus on training efficiency: MgL predominantly operates during the training phase rather than at runtime. While this design leverages mainstream GPU acceleration (e.g., CUDA operators) for offline or batch training, it may not align perfectly with edge-deployment scenarios demanding continuous model updates under limited hardware resources.

3) Robustness to diverse conditions: Although we evaluate on five datasets under varying noise ratios (up to 90%), real-world scenarios can involve more complex issues such as extreme class imbalance, dynamic fault evolution, or multi-sensor data fusion. Integrating adaptive mechanisms (e.g., online learning or incremental updates) could improve MgL's robustness across a broader range of operational contexts.

4) Choice of DSmT: MgL selected a DSmT-based evidence-fusion scheme in a real-valued setting to ensure compatibility with existing deep learning toolchains and straightforward integration into classical GPU-based training. However, the transferable belief model (TBM) [35] could offer additional benefits if combined with genuine quantum computing—potentially enabling more efficient parallel operations and natural handling of amplitude-based calculations in edge-deployment or large-scale industrial scenarios where quantum hardware is viable. Such a TBM-based approach might open new directions for bridging complex amplitudes with classical gradient updates, though at present, this entails higher implementation complexity and specialized hardware support [36]. Future research along these lines may prove valuable if quantum computing achieves broader practical adoption in industrial fault diagnosis.

V. Conclusion

Motivated by the challenge of "*How can the confirmation bias caused by inaccurate predictions in LNL processes be mitigated to enhance the generalization capability of diagnosis* *models in high-noise environments*", we draw inspiration from the uncertainty in quantum collapse and integrate multi-source evidence fusion to propose a **M**ulti-**g**ranularity **L**abeling (*MgL*) approach. MgL constructs three tiers of granularity in the labels, thereby significantly improving the diagnostic model's tolerance to label noise. Extensive experiments demonstrate that, particularly under a high noise rate of $\eta = 90\%$, MgL surpasses existing methods by over 4.20% to 55.38% in accuracy. Such robust performance enables high-precision diagnosis even on datasets with severely noisy labels, which is expected to improve the automation and intelligence level of related industrial fields by accessing more non manually annotated datasets with lower annotation costs.

References

[1] H. Fang, J. Xiang, J. An, H. Liu, H. Li, Y. Cui, and F. Dunkin, "Fcdg: A central dogma-inspired approach for cross-domain fault diagnosis," *IEEE Sensors Journal*, vol. 25, no. 3, pp. 5192–5199, 2025.

[2] C. Wang, Z. Wang, H. Dong, S. Lauria, W. Liu, Y. Wang, F. Fadzil, and X. Liu, "Fusionformer: A novel adversarial transformer utilizing fusion attention for multivariate anomaly detection," *IEEE Transactions on Neural Networks and Learning Systems*, pp. 1–14, 2025.

[3] C. Wang, Z. Wang, Q. Liu, H. Dong, and W. Sheng, "Support-sample-assisted domain generalization via attacks and defenses: Concepts, algorithms, and applications to pipeline fault diagnosis," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 4, pp. 6413–6423, 2024.

[4] C. Wang, Z. Wang, H. Liu, H. Dong, and G. Lu, "An optimal unsupervised domain adaptation approach with applications to pipeline fault diagnosis: Balancing invariance and variance," *IEEE Transactions on Industrial Informatics*, vol. 20, no. 8, pp. 10 019–10 030, 2024.

[5] F. Dunkin, X. Li, H. Li, G. Wu, C. Hu, and S. S. Ge, "Mgcnl: A sample separation approach via multi-granularity balls for fault diagnosis with the interference of noisy labels," *IEEE Transactions on Automation Science and Engineering*, vol. 22, pp. 7748–7761, 2025.

[6] F. Dunkin, X. Li, C. Hu, G. Wu, H. Li, X. Lu, and Z. Zhang, "Like draws to like: A multi-granularity ball-intra fusion approach for fault diagnosis models to resists misleading by noisy labels," *Advanced Engineering Informatics*, vol. 60, p. 102425, 2024.

[7] J. Zhang, V. S. Sheng, T. Li, and X. Wu, "Improving crowdsourced label quality using noise correction," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 29, no. 5, pp. 1675–1688, 2018.

[8] M. Liang, J.-C. Su, S. Schulter, S. Garg, S. Zhao, Y. Wu, and M. Chandraker, "Aide: An automatic data engine for object detection in autonomous driving," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 14 695–14 706.

[9] N.-r. Kim, J.-S. Lee, and J.-H. Lee, "Learning with structural labels for learning with noisy labels," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2024, pp. 27 610–27 620.

[10] C. Zhang, S. Bengio, M. Hardt, B. Recht, and O. Vinyals, "Understanding deep learning (still) requires rethinking generalization," *Communications of the ACM*, vol. 64, no. 3, pp. 107–115, 2021.

[11] H. Song, M. Kim, D. Park, Y. Shin, and J.-G. Lee, "Learning from noisy labels with deep neural networks: A survey," *IEEE Transactions on Neural Networks and Learning Systems*, vol. 34, no. 11, pp. 8135–8153, 2023.

[12] X. Zhou, X. Liu, D. Zhai, J. Jiang, and X. Ji, "Asymmetric loss functions for noise-tolerant learning: Theory and applications," *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 45, no. 7, pp. 8094–8109, 2023.

[13] F. R. Cordeiro and G. Carneiro, "Anne: Adaptive nearest neighbours and eigenvector-based sample selection for robust learning with noisy labels," *Pattern Recognition*, vol. 159, p. 111132, 2025.

[14] S. Zheng, P. Wu, A. Goswami, M. Goswami, D. Metaxas, and C. Chen, "Error-bounded correction of noisy labels," in *International Conference on Machine Learning*. PMLR, 2020, pp. 11 447–11 457.

[15] Y. Wang, X. Ma, Z. Chen, Y. Luo, J. Yi, and J. Bailey, "Symmetric cross entropy for robust learning with noisy labels," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2019, pp. 322–330.

[16] H. Wei, L. Feng, X. Chen, and B. An, "Combating noisy labels by agreement: A joint training method with co-regularization," in *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, 2020, pp. 13 726–13 735.

[17] X. Xia, B. Han, Y. Zhan, J. Yu, M. Gong, C. Gong, and T. Liu, "Combating noisy labels with sample selection by mining high-discrepancy examples," in *Proceedings of the IEEE/CVF international conference on computer vision*, 2023, pp. 1833–1843.

[18] M. Chen, Y. Zhao, B. He, Z. Han, J. Huang, B. Wu, and J. Yao, "Learning with noisy labels over imbalanced subpopulations," *IEEE Transactions on Neural Networks and Learning Systems*, pp. 1–12, 2024.

[19] Y. Lu and W. He, "Mitigating noisy supervision using synthetic samples with soft labels," *ArXiv*, vol. abs/2406.16966, 2024.

[20] M. Kashiwagi, K. Maeda, R. Togo, T. Ogawa, and M. Haseyama, "Enhancing noisy label learning via unsupervised contrastive loss with label correction based on prior knowledge," *ICASSP 2024 - 2024 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pp. 6235–6239, 2024.

[21] H. Wang, C. Li, P. Ding, S. Li, T. Li, C. Liu, X. Zhang, and Z. Hong, "A novel transformer-based few-shot learning method for intelligent fault diagnosis with noisy labels under varying working conditions," *Reliability Engineering & System Safety*, vol. 251, p. 110400, 2024.

[22] F. Li, K. Li, J. Tian, and J. Zhou, "Regroup median loss for combating label noise," in *Proceedings of the AAAI Conference on Artificial Intelligence*, vol. 38, no. 12, 2024, pp. 13 474–13 482.

[23] J. Wei, H. Liu, T. Liu, G. Niu, M. Sugiyama, and Y. Liu, "To smooth or not? when label smoothing meets noisy labels," in *International Conference on Machine Learning*. PMLR, 2022, pp. 23 589–23 614.

[24] M. Sheng, Z. Sun, T. Chen, S. Pang, Y. Wang, and Y. Yao, "Foster adaptivity and balance in learning with noisy labels," in *European Conference on Computer Vision*. Springer, 2025, pp. 217–235.

[25] Y. Li, H. Han, S. Shan, and X. Chen, "Disc: Learning from noisy labels via dynamic instance-specific selection and correction," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, 2023, pp. 24 070–24 079.

[26] K. Camilleri, "Constructing the myth of the copenhagen interpretation," *Perspectives on Science*, vol. 17, no. 1, pp. 26–57, 05 2009. [Online]. Available: https://doi.org/10.1162/posc.2009.17.1.26

[27] M. Cramer, M. B. Plenio, S. T. Flammia, R. Somma, D. Gross, S. D. Bartlett, O. Landon-Cardinal, D. Poulin, and Y.-K. Liu, "Efficient quantum state tomography," *Nature communications*, vol. 1, no. 1, p. 149, 2010.

[28] L. Xinde, F. Dunkin, and J. Dezert, "Multi-source information fusion: Progress and future," *Chinese Journal of Aeronautics*, vol. 37, no. 7, pp. 24–58, 2024.

[29] W. A. Smith and R. B. Randall, "Rolling element bearing diagnostics using the case western reserve university data: A benchmark study," *Mechanical Systems and Signal Processing*, vol. 64-65, pp. 100–131, 2015.

[30] T. Li, Z. Zhou, S. Li, C. Sun, R. Yan, and X. Chen, "The emerging graph neural networks for intelligent fault diagnostics and prognostics: A guideline and a benchmark study," *Mechanical Systems and Signal Processing*, vol. 168, p. 108653, 2022.

[31] H. Fang, J. Deng, D. Chen, W. Jiang, S. Shao, M. Tang, and J. Liu, "You can get smaller: A lightweight self-activation convolution unit modified by transformer for fault diagnosis," *Advanced Engineering Informatics*, vol. 55, p. 101890, 2023.

[32] J. Deng, H. Liu, H. Fang, S. Shao, D. Wang, Y. Hou, D. Chen, and M. Tang, "Mgnet: A fault diagnosis approach for multi-bearing system based on auxiliary bearing and multi-granularity information fusion," *Mechanical Systems and Signal Processing*, vol. 193, p. 110253, 2023.

[33] J. Li, C. Xiong, and S. C. Hoi, "Learning from noisy data with robust representation learning," in *Proceedings of the IEEE/CVF International Conference on Computer Vision*, 2021, pp. 9485–9494.

[34] A. Iscen, J. Valmadre, A. Arnab, and C. Schmid, "Learning with neighbor consistency for noisy labels," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, June 2022, pp. 4672–4681.

[35] Z. Liu, F. Xiao, C.-T. Lin, and Z. Cao, "A robust evidential multisource data fusion approach based on cooperative game theory and its application in eeg," *IEEE Transactions on Systems, Man, and Cybernetics: Systems*, vol. 54, no. 2, pp. 729–740, 2024.

[36] Z. Guo, Z. Wan, Q. Zhang, X. Zhao, Q. Zhang, L. M. Kaplan, A. Jøsang, D. H. Jeong, F. Chen, and J.-H. Cho, "A survey on uncertainty reasoning and quantification in belief theory and its application to deep learning," *Information Fusion*, vol. 101, p. 101987, 2024.

Fir Dunkin (Graduate Student Member, IEEE) received the M.E. degree from the School of Automation Engineering, Northeast Electric Power University, Jilin, China, in 2022.

Currently, he is pursuing the Ph.D. degree with the Southeast University, and his research interests include information fusion, uncertainty reasoning, and mechatronics etc. Meanwhile, he has published over 20 peer-reviewed papers, including 4 ESI highly cited papers, and submitted over 150 peer-reviewed results as a reviewer for journals including IEEE TII/TFS/TCSVT/TMech/TIM, InfFus, MSSP, PR, ESWA, KBS, EAAI, NeuNet, etc.

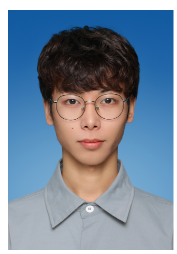
Xinde Li (Senior Member, IEEE) received his Ph.D. degree in control theory and control engineering from the Department of Control Science and Engineering, Huazhong University of Science and Technology, Wuhan, China, in 2007. After that, he joined the School of Automation, Southeast University, Nanjing, China, where he is currently a professor. He was elected as a senior member of Institute of Electrical and Electronics Engineers in 2016, an academician with the Russian Academy of Natural Sciences and a cover person of Scientific Chinese in 2021 and a fellow of the Institution of Engineering and Technology in 2022. He is the vice director of Intelligent Robot Committee of the Chinese Association for Artificial Intelligence from 2017, the vice director of Intelligent Products and Industry Working Committee of the Chinese Association for Artificial Intelligence from 2019 and the vice director of Intelligent Manufacturing Systems & Technology Professional Committee of Chinese Association of Automation from 2022.

His research interests include intelligent robot, and unmanned system, human-machine interaction, computer vision, information fusion. He was the principal of about 50 projects including key and major projects of NSFC, etc. He published more than 100 papers, 5 books and won 32 national invention patents. He also won many prizes including global award for outstanding scientist in artificial intelligence and robotics, Asia award for outstanding researcher, gold award in Geneva international invention, etc. He was also the associate editors of several reputative international journals, i.e. IEEE Transactions on Fuzzy Systems, Journal of Systems Engineering and Electronics, etc.

Zhentong Zhang received the B.S. degree in electrical engineering and automation from the School of IoT Engineer ing, Jiangnan University, Wuxi, China, in 2016, and the M.S. degree in control engineering from the School of Electrical Engineering, Xinjiang Uni versity, Ürümqi, China, in 2021. He is currently pursuing the Ph.D. degree in electronic information engineering with the School of Cyber Science and Engineering, Southeast University, Nanjing, China. His research interests include the computer vision, deep learning, and unmanned system security.

Kui Wang , 1991, is a senior engineer in the Simulation System R&D Department of North Information Control Group Co., Ltd. His main research areas include the construction technology and application of digital intelligence battlefield, digital twin technology and application, and collaborative training technology and application with and without human presence.

Tianrong Gao , 1985, Deputy Chief Engineer and Senior Engineer of North Information Control Group Co., Ltd., holds a PhD and doctoral supervisor. Her main research areas include virtual battlefield simulation, LVC collaborative simulation, and complex system integration.



Guoliang Wu received the B.E. degree from the School of Information Science and Technology, Beijing University of Chemical Technology, Beijing, China, in 2020.

At present, he is pursuing the Ph.D. degree with the School of Automation, Southeast University, Nanjing, China. His current research interests include Multi-granular information fusion, Intelligent Game, Neural networks, and Pattern recognition etc.

Zhijun Li (Fellow IEEE) received the Ph.D. degree in mechatronics from the Shanghai Jiao Tong University, Shanghai, China, in 2002. From 2003 to 2006, he was a Post-Doctoral Fellow with The University of Electro Communications, Tokyo, Japan; and the National University of Singapore, Singapore. He is currently a Chair Professor with Tongji University, Shanghai, where he has been the Dean of the School of Mechanical Engineering, since 2024. He has published over 400 papers, where the prestigious contributions were wearable robotics and bio-mechatronics systems. His current research interests include wearable robotics, bio-mechatronics systems, and intelligent control. He is a fellow of AAIA. He has received the Distinguished Lecturer (RAS), the Web of Science Highly Cited Researcher (2019-2022), the 2018 National Ten-Thousand Talents Program in China, and the 2016 National Distinguished Young Scholar (NSFC).